Affine Monodromy and a Focus-Focus System

Donghae Lee

Februrary 2026

Abstract

If a Lagrangian torus fibration admits a singular fiber, the mirror may need to be constructed by gluing several affine torus charts, and the integral affine structure on its base may develop nontrivial monodromy. In this note, I begin by briefly discussing SYZ mirror symmetry for toric varieties and introduce the focus-focus singularity, which generates nontrivial affine monodromy in the base of a Lagrangian torus fibration. Then I introduce Auroux's system as an example and show how the inconsistency caused by affine monodromy is compensated by wall-crossing phenomena.

Contents

1 Focus-Focus Singularities **1**

2 SYZ for Toric Varieties and Wall-Crossing **2**

3 Gluing Affine Charts and Instanton Correction **5**

1 Focus-Focus Singularities

First, I define a *focus-focus singularity*, a type of singularity that induces nontrivial monodromy. The following definitions follow Chapter 6 of [Eva21].

Definition 1.1. In the integrable system $(\mathbb{R}^4, dx_1 \wedge dx_2 + dy_1 \wedge dy_2)$, consider a pair of Hamiltonians

$$H_1 = x_1x_2 - y_1y_2, H_2 = y_1x_2 - x_1y_2. \tag{1}$$

Using complex coordinates $z_1 = x_1 + iy_1, z_2 = x_2 + iy_2$, this system can be written as $H := H_1 + iH_2 = -\bar{z}_1 z_2$. It has a singularity at $(0,0)$, and we refer to this system as the *standard focus-focus system*

Definition 1.2. Let $H : X \to \mathbb{R}^2$ be an integrable system. If there exists a Darboux chart φ near a point $x \in X$ such that $H \circ \varphi$ is the standard focus-focus system, then we call x a *focus-focus singularity*.

If x is a focus-focus singularity, the fiber of H containing x looks like a pinched torus, and fibers near x are smooth tori. Thus, the standard focus-focus system serves as a model of a torus fibration with a degenerated singular fiber.

A focus-focus singularity has nontrivial monodromy around it; this is the reason why it causes a problem in the SYZ mirror construction later.

Proposition 1.3. *For a generic torus fiber $H^{-1}(c)$ ($c \neq 0$) of a standard focus-focus system, let α be a vanishing cycle and β be a complementary cycle to α such that (α, β) forms a basis of $\pi^1(H^{-1}(c))$. Then the monodromy around the critical value sends α to α and β to $\alpha + \beta$. Using the ordered basis β, α, this corresponds to the shear matrix $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$.*

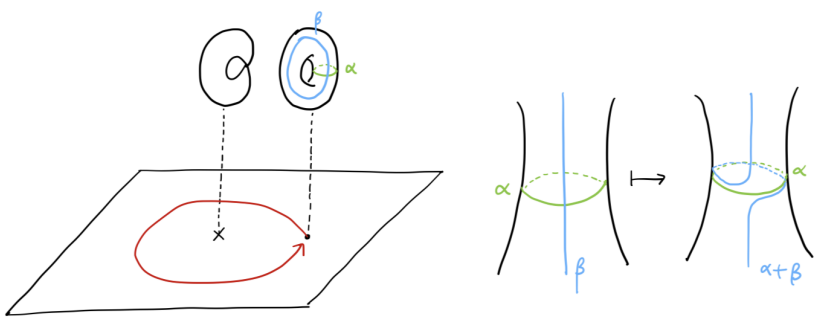

Figure 1: Monodromy of a focus-focus singularity.

Now I introduce a representative example of a Lagrangian torus fibration with a focus-focus singularity—this is an example from Chapter 5 in [Aur07].

Example 1.4. Consider $\mathbb{C}^2$ with the standard symplectic form and an integrable system $H := (H_1, H_2) : \mathbb{C}^2 \to \mathbb{R}^2$, consisting of a pair of Hamiltonians $H_1(z_1, z_2) = |z_1 z_2 - \epsilon|^2$ and $H_2(z_1, z_2) = \frac{1}{2}(|z_1|^2 - |z_2|^2)$. Since H_1 and H_2 Poisson commute, a fiber $H^{-1}(r^2, \frac{1}{2}\lambda)$ is Lagrangian for $r > 0$ and $\lambda \in \mathbb{R}$. Denote this Lagrangian torus fiber by $T_{\gamma(r),\lambda}$. Also, H has a critical point at $(0, 0)$, which is a focus-focus singularity. (This can be verified by computation using two local complex coordinates (x, y) with a chart $z_1 = tx + \sqrt{\epsilon}$ and $z_2 = ty + \sqrt{\epsilon}$.)

To give a torus fibration structure on Example 1.4, consider the conic fibration $\pi : \mathbb{C}^2 \to \mathbb{C}$ given by $\pi(z_1, z_2) = z_1 z_2$. It has a singular fiber at $0 \in \mathbb{C}$. Figure 2 shows how the torus $T_{\gamma(r),\lambda}$ can be described in terms of this conic fibration. A torus $T_{\gamma(r),\lambda}$ is obtained by parallel transporting a circle of height λ in a conic fiber along a circle $\gamma(r)$ of radius r centered at ϵ. This yields a Lagrangian torus fibration of $\mathbb{C}^2 \setminus \{xy - \epsilon = 0\}$. Moreover, the fiber $T_{\gamma(|\epsilon|),0}$, which contains the point $(0, 0)$, is homeomorphic to a pinched torus, indicating that $(0, 0)$ is a focus-focus singularity.

2 SYZ for Toric Varieties and Wall-Crossing

Before defining the wall-crossing phenomenon, I briefly review the concept of SYZ mirror symmetry for toric varieties. Let X be a toric variety with moment map $\mu : X \to \mathbb{R}^n$. Denote the image of

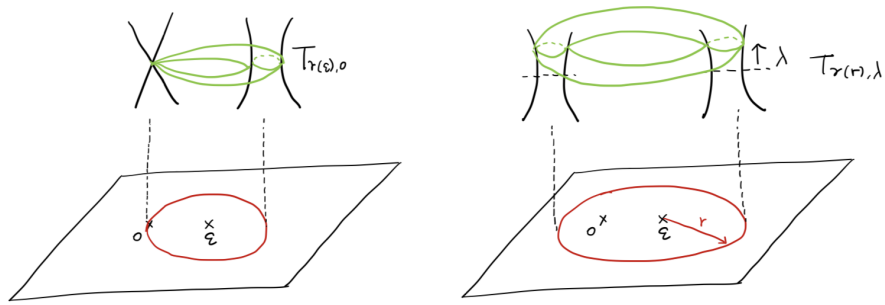

Figure 2: Lagrangian torus fibration given by $T_{\gamma(r),\lambda}$.

the moment map μ by $\Delta \subset \mathbb{R}^n$. For each $b \in \mathrm{Int}(\Delta)$, we can dualize the torus fiber $L_b = \mu^{-1}(b)$ to $(b, \triangledown)$ in the language of T-duality. Since $\mathrm{Int}(\Delta) \simeq \mathbb{R}^n_{>0}$, these fibers are dualized to $(\mathbb{C}^*)^n$.

However, we lose the information about the degeneration of the torus along the boundary of Δ. We need to add extra data to record this degeneration—it is encoded by a function called the *superpotential* $W : (\mathbb{C}^*)^n \to \Lambda_{\mathbf{k}}$.

Definition 2.1. A *superpotential* W is a function defined as follows:

$$W((b, \triangledown)) = \sum_{\beta \in \pi^2(M, L_b), \mu(\beta)=2} n_\beta \cdot T^{-\omega(\beta)} \triangledown(\partial\beta) \in \Lambda_{\mathbf{k}} \tag{2}$$

For each relative homotopy class $\beta \in \pi^2(M, L_b)$, n_β denotes the number of holomorphic discs in the class β. The term $T^{-\omega(\beta)}$ is a Novikov weight recording the symplectic area of holomorphic discs, and the final factor $\triangledown(\partial\beta)$ keeps track of the holonomy along the boundary of a disc in the class β. We denote the latter two terms $T^{-\omega(\beta)} \triangledown(\partial\beta)$ by a new variable z_β for each β.

Proposition 2.2. *A toric variety X admits a mirror given by the pair $((\mathbb{C}^*)^n, W)$; we refer to this mirror as the Landau–Ginzburg mirror.*

Note that a superpotential is defined by counting Maslov index 2 discs. However, it is usually defined under the assumption that there is no Lagrangian fiber that bounds Maslov index 0 discs. If such discs exist, a *wall-crossing phenomenon* appears, and the superpotential may fail to be well defined.

Definition 2.3. For a Lagrangian fiber L, the set of points $p \in L$ that lie on the boundary of a Maslov index 2 disc which meets, at the boundary, another Maslov index 0 disc forms a codimension 1 subset $\mathcal{W} \subset L$. We call $\mathcal{W}$ a *wall*, and it separates L into various chambers, each of which has a different value of the superpotential.

The wall causes a discontinuity of the superpotential W at b. Let L_t for $t \in [-\epsilon, \epsilon]$ be a continuous family of Lagrangian fibers, and suppose that L_0 bounds a disc u_α in the class $\alpha \in \pi^2(M, L_0)$ with $\mu(\alpha) = 0$. Then the disc counts $n_\beta(L_t)$ vary as t crosses 0. This variation in disc counting makes W discontinuous.

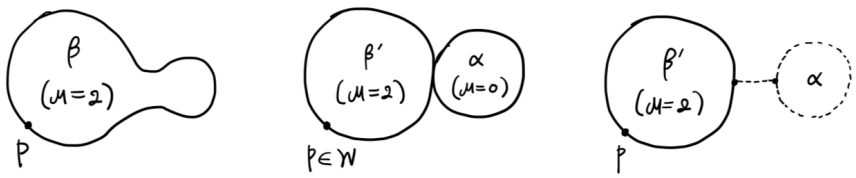

Figure 3: Description of wall-crossing. (same figure with [Aur07])

Remark 2.4. From the above discussion, a wall $\mathcal{W}$ of a single fiber L also induces a wall in the base. The base can be divided along the image of Lagrangian fibers that bound Maslov index zero discs. Each chamber corresponds to a single affine chart of its mirror, and each affine chart has a superpotential defined as a Laurent polynomial.

Let the superpotential in the chamber where a disc in the class β' and a disc in the class α are separated (the right side in Figure 3) be denoted by W^-, and the superpotential in the chamber where a disc in the class α is glued to a disc in the class β' (the left side in Figure 3) be denoted by W^+. Then we can describe the difference between these two superpotentials using the disc counting result for the disc u_α and its multiple covers. For a clear convention, define the sign of t for L_t so that L_ϵ is contained in the chamber with superpotential W^+ and $L_{-\epsilon}$ is contained in the chamber with superpotential W^-.

Definition 2.5. For homotopy classes $\beta \in \pi^2(M, L)$ with $\mu(\beta) = 2$ and $\alpha \in \pi^2(M, L)$ with $\mu(\alpha) = 0$ that have nonzero intersection number $[\partial\beta] \cdot [\partial\alpha]$, denote by $n_{\beta+m\alpha}$ the number of holomorphic discs in the homotopy class $\beta + m\alpha$ with boundary on L. Then the *generating series* F_t for the given β, α, and the family of Lagrangian fibers L_t is defined as follows:

$$F_t(q) := \sum_{m\in\mathbb{Z}} n_{\beta+m\alpha}(L_t)q^m \tag{3}$$

Now, denote $n_{m\alpha}$ similarly to $n_{\beta+m\alpha}$ as the number of holomorphic discs that are m-fold covers of u_α. Then we observe that

$$F_\epsilon(q) - F_{-\epsilon}(q) = h_\alpha(q)^{[\partial\beta]\cdot[\partial\alpha]} \tag{4}$$

where $h_\alpha(q) = 1 + n_\alpha q + 2n_{2\alpha}q^2 + \cdots$.

By substituting q with z_α, we obtain the contribution of the glued discs as in the left side of Figure 3; this contribution is $z_\beta h_\alpha(z_\alpha)^{[\partial\beta]\cdot[\partial\alpha]}$. The above discussion can be summarized in Proposition 2.6.

Proposition 2.6. *The two superpotentials from the respective affine charts, W^+ and W^-, are related by a holomorphic change of variables*

$$z_\beta \mapsto z_\beta h(z_\alpha)^{[\partial\beta]\cdot[\partial\alpha]} \tag{5}$$

for each $\beta \in \pi^2(M, L_0)$.

Remark 2.7. Equation 5 gives a holomorphic coordinate transformation between two affine charts across the wall. (We will check this for Example 1.4 in Section 3.)

In summary, a superpotential is defined by counting Maslov index 2 discs, in order to record the torus degeneration in the SYZ mirror of toric varieties. If some fiber bounds Maslov index

zero discs, then the superpotential cannot be defined globally, and the discrepancy of W across a wall crossing is given by a coordinate change whose exponent is determined by the algebraic intersection number $[\partial\alpha] \cdot [\partial\beta]$, for α with $\mu(\alpha) = 0$ and each β with $\mu(\beta) = 2$.

3 Gluing Affine Charts and Instanton Correction

However, a focus-focus singularity has nontrivial affine monodromy, as seen in Proposition 1.3, and this causes a *failure* of consistency when gluing the affine structure to construct a mirror.

Let us consider Example 1.4 and its Lagrangian torus fiber $T_{\gamma(r),\lambda}$ again. Example 1.4 is constructed on $\mathbb{C}^2 \setminus xy - \epsilon = 0$, but from now on we replace it by $\mathbb{CP}^2 \setminus D$ for the divisor $D = \{Z_0(Z_1Z_2 - \epsilon Z_0^2)\}$, which denotes the union of the line at infinity and the fiber $\{xy - \epsilon\}$.

Remark 3.1. The reason we replace $\mathbb{C}^2 \setminus \{xy - \epsilon = 0\}$ by $\mathbb{CP}^2 \setminus D$ is that D is an anticanonical divisor. This shows that Example 1.4 is a Calabi–Yau variety equipped with the holomorphic volume form $\Omega = \frac{dz_1 \wedge dz_2}{z_1 z_2 - \epsilon}$ and the Fubini–Study form ω_{FS} as a symplectic form. This fact is not used in this note, but another advantage of this description will appear later. (See Figure 4.)

To identify where the wall lies in $\mathbb{CP}^2 \setminus D$, we need to determine which fibers bound Maslov index zero discs.

Lemma 3.2. *A Lagrangian torus $T_{\gamma(r),\lambda}$ bounds a Maslov index zero disc if and only if $r = |\epsilon|$.*

Lemma 3.2 tells us that wall-crossing occurs at $r = |\epsilon|$ in the base. We now compute the superpotential in the two chambers $r > |\epsilon|$ and $r < |\epsilon|$, respectively.

First, consider the case $r > |\epsilon|$. To count Maslov index 2 discs bounded by $T_{\gamma(r),\lambda}$, we deform the circle $\gamma(r)$ to a small circle centered at the origin in the base (see Figure 5). This deformation does not change the holomorphic discs bounded by this torus, and it also does not change their homotopy classes. The deformed torus fiber consists of the points (z_1, z_2) satisfying $|z_1|^2 + |z_2|^2 = r^2$ and $\frac{1}{2}(|z_1|^2 - |z_2|^2) = \lambda$. Thus, after deformation, the Lagrangian torus becomes $S^1(r_1) \times S^1(r_2) \subset \mathbb{C}^2$ for fixed r_1 and r_2.

From the above discussion, we can depict three Maslov index 2 discs as in Figure 4 and obtain a superpotential as in Proposition 3.3. Even though $\mathbb{CP}^2 \setminus D$ is not a toric variety, since there is no Maslov index zero disc before crossing the wall, we can draw discs using the moment map of $\mathbb{CP}^2$.

In Figure 4, discs in the classes β_1 and β_2 are sections of the conic fibration over $\gamma(r)$ and its interior: a disc in the class β_1 meets $\pi^{-1}(0)$ along the x-axis, and a disc in the class β_2 meets $\pi^{-1}(0)$ along the y-axis. Moreover, let H be a generator of $H^*(\mathbb{CP}^2)$, which is a sphere class with Maslov index 6 and area $\omega(H) = \Lambda$. Since the Maslov index is additive with respect to the homotopy class, we have $\mu(H - \beta_1 - \beta_2) = 2$, and it corresponds to the term $\frac{T^{-\Lambda}}{z_{\beta_1} z_{\beta_2}}$ in the superpotential.

Proposition 3.3. *For $r > |\epsilon|$, the superpotential is*

$$W^- = x + y + \frac{T^{-\Lambda}}{xy} \tag{6}$$

for variables x and y corresponding to the homotopy classes β_1 and β_2, respectively.

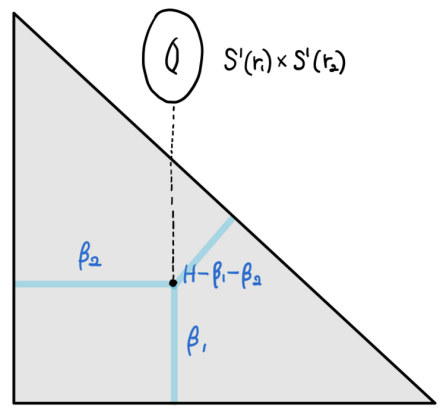

Figure 4: Three Maslov index 2 discs in the homotopy classes β_1, β_2, and $H - \beta_1 - \beta_2$.

Remark 3.4. A more subtle discussion is needed to explain why these discs can be described on the image of the moment map of $\mathbb{CP}^2$, even though the anticanonical divisor D in Example 1.4 does not coincide with a toric divisor; however, I omit an explicit explanation.

Now consider the case $r < |\epsilon|$. In this case, there is only one homotopy class of holomorphic disc that is a section of π, but there is also a Maslov index zero disc constructed as a Lefschetz thimble (its projection under π is a path connecting the origin and a point on $\gamma(r)$). Denote the corresponding homotopy classes by β and α, respectively. In the right side of Figure 5, the disc with blue boundary denotes a disc in the class β, and the disc with pink boundary denotes a disc in the class α. As before, there is a sphere class H with Maslov index 6. By the additivity of the Maslov index, a Maslov index 2 disc class should be of the form $\beta + k(H - 3\beta) + m\alpha$. Checkanov and Schlenk proved that the only classes that can occur are β and $H - 2\beta + k\alpha$ for $k \in \{-1, 0, 1\}$, and they also computed the numbers of discs in these classes. (I omit the proof of this result, which involves an explicit construction of holomorphic discs.) Their result yields Proposition 3.5.

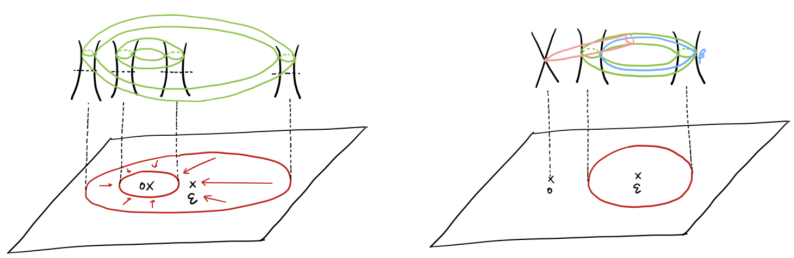

Figure 5: Description of the Maslov index 2 disc counts. (left: $r > |\epsilon|$, right: $r < |\epsilon|$)

Proposition 3.5. *For $r < |\epsilon|$, the superpotential is*

$$W^+ = u + \frac{T^{-\Lambda}}{u^2 w} + 2\frac{T^{-\Lambda}}{u^2} + \frac{T^{-\Lambda} w}{u^2} = u + \frac{T^{-\Lambda}(1+w)^2}{u^2 w} \tag{7}$$

for variables u and w corresponding to the homotopy classes β and α, respectively.

I now conclude this note by explaining the discrepancy that arises when we glue these two affine charts with their superpotentials, and how it can be resolved by *disc instanton correction*.

Before computing the explicit coordinate transitions, observe that $W^-(x,y)$ and $W^+(u,w)$ are related by the change of variables $u = x + y$ and $w = x/y$. However, we obtain different coordinate transition maps depending on which side of the wall we cross, namely whether $\lambda > 0$ or $\lambda < 0$.

When we cross the wall along ① in Figure 6 (that is, when $\lambda > 0$), the homotopy classes of the discs change as follows, corresponding to the following change of variables:

$$\begin{cases} \alpha \leftrightarrow \beta_1 - \beta_2 & w \leftrightarrow x/y \\ \beta \leftrightarrow \beta_2 & u \leftrightarrow y \\ H - 2\beta - \alpha \leftrightarrow H - \beta_1 - \beta_2 & T^{-\Lambda}/u^2 w \leftrightarrow T^{-\Lambda}/xy \end{cases} \tag{8}$$

Now consider the case of crossing the wall along ② in Figure 6 (that is, when $\lambda < 0$). We can compute the change of variables in the same way as in the above case:

$$\begin{cases} -\alpha \leftrightarrow \beta_2 - \beta_1 & w \leftrightarrow y/x \\ \beta \leftrightarrow \beta_1 & u \leftrightarrow x \\ H - 2\beta + \alpha \leftrightarrow H - \beta_1 - \beta_2 & T^{-\Lambda} w/u^2 \leftrightarrow T^{-\Lambda}/xy \end{cases} \tag{9}$$

Remark 3.6. The difference in the sign of α in Equations 8 and 9 comes from the difference in orientation. The relative homotopy class of the Maslov index 0 discs that cause the wall crossing is the same on the two parts of the wall, but the induced orientations are opposite.

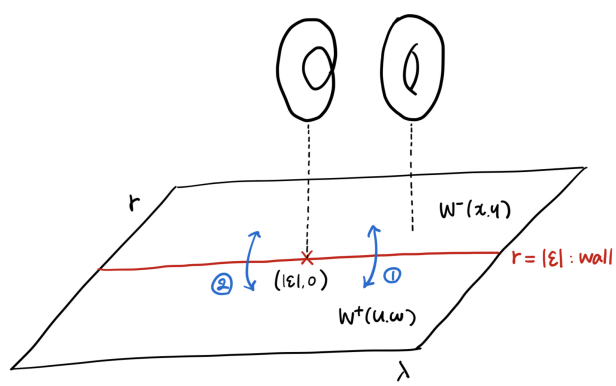

Figure 6: Glued SYZ mirror obtained by using two affine charts.

Consider rotating around the singular fiber and returning to the affine chart with coordinates (u,w), crossing the wall ② first and then crossing the wall ①. Under this loop, the coordinate (u,w) is sent to $(x, x/y)$ and then returns to (uw, w). In terms of the basis $(\log u, \log w)$, this change of variables corresponds to the shear matrix $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, which coincides with the monodromy appearing in Proposition 1.3.

To correct this discrepancy, we need to add the *instanton correction* $u \mapsto u(1+w)$ on the $\lambda > 0$ part of the wall and $u \mapsto u(1+w^{-1})$ on the $\lambda < 0$ part of the wall. After applying these corrections, we

obtain the change of variables $u \leftrightarrow x + y$ and $w \leftrightarrow x/y$, as expected. This allows us to glue the two superpotentials W^- and W^+, yielding a global superpotential on $(\mathbb{C}^*)^2$.

Remark 3.7. To compute the instanton correction explicitly using Equation 5, we would need to count the moduli space of Maslov index zero discs and its multiple covers. However, achieving regularity for these moduli spaces is very difficult, and typically requires techniques such as Kuranishi charts. Instead, a more effective approach is to guess an instanton correction that yields a global superpotential, based on the fact that we *can* in principle count the moduli of Maslov index zero discs, and then recover the numbers $n_{m\alpha}$ from the instanton correction formula. This provides a clearer way to understand the phenomenon.

References

[1] D. Auroux, *Mirror Symmetry and T-duality in the Complement of an Anticanonical Divisor*, J. Gökova Geom. Topol. (2007)

[2] J. Evans, *Lectures on Lagrangian Torus Fibrations*, LMS Student Texts 105, Cambridge University Press. (2023)